MILITARY SPECIFICATION

PUDDINGS; CHOCOLATE, CHOCOLATE PEPPERMINT, AND VANILLA, POWDER (FOR ORAL LIQUID FEEDING)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers chocolate, chocolate peppermint, and vanilla pudding powders for oral liquid feeding. This product is one component of a meal that, when consumed with other components in three meals or supplementary nourishments, provides a nutritionally adequate, orally consumed diet.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-P-44151	- Packaging and Meal Assembly of Powder (For Oral Liquid Feeding)

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Grade A Pasteurized Milk Ordinance

(Application for copies should be addressed to the Department of Health and Human Services, U.S. Public Health Service, Food and Drug Administration, Milk Safety Branch, Code HFF-346, 200 C Street S.W., Washington, DC 20204.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to the Dairy Standardization Section, Dairy Division, Room 2750-So., Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Margarine. The margarine shall conform to the Federal Food, Drug, and Cosmetic Act Standards of Identity, 21 CFR 166.40 and 166.110. The margarine shall have a sweet pleasing taste and aroma, free from rancidity or other objectionable flavors and odors. It shall have a medium yellow color and a smooth body and texture free from crumbliness, graininess, grittiness, oiliness, or droplets of free moisture. The margarine shall be salted. The margarine shall be in excellent condition, free from mold, slime, or discoloration. The margarine shall be manufactured in a plant listed in the Dairy Plants Surveyed and Approved for USDA Grading Service.

3.2.2 Milk. Milk shall be Grade A pasteurized, homogenized milk that complies with requirements of the Grade A Pasteurized Milk Ordinance of the U.S. Public Health Service and 21 CFR 131.110. The milk shall have a clean, pleasing flavor, but may have the following flavors to a slight degree: cooked (as appropriate for the pasteurization process), feed, or flat. It shall be free from all other objectionable flavors.

3.2.3 Cream, heavy. The heavy cream shall possess a sweet, pleasing, and desirable flavor. It shall be homogenized and free from objectionable tastes and odors. The cream shall possess a uniform white to light-cream color. The product shall comply with the composition requirements specified in 21 CFR 131.150.

3.2.4 Sugar, granulated. Sugar shall be clean, refined sucrose, granulated cane or beet, or a combination of both.

3.2.5 Cocoa. The cocoa powder shall conform to the Federal Food, Drug, and Cosmetic Act Standards of Identity for medium fat dutched cocoa (21 CFR 163.113). When washed with petroleum ether, not less than 98 percent by weight shall pass through a U.S. Standard No. 200 sieve. The pH shall be between 7 and 8.5.

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3.2.6 Peppermint extract. Peppermint oil in the extract shall be of the natural plant, Mentha piperita origen. The extract shall contain oil of peppermint, water, and not less than 89 percent alcohol by volume. The extract shall possess a flavor and aroma characteristic of peppermint.

3.2.7 Salt. Salt shall be iodized, clear white, refined sodium chloride, with or without anticaking agents.

3.2.8 Vanillin. Vanillin shall be a white powder with a characteristic vanillin odor. It shall be the purity specified in the Food Chemicals Codex.

3.2.9 Eggs, whole. Whole eggs may be liquid or frozen. They shall be processed in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The whole eggs shall be egg yolks and egg whites in their natural proportions as broken directly from the shell as evidenced by a USDA Egg Products Inspection Certificate. The whole eggs shall contain no additives and shall be in satisfactory condition, free of objectionable odors, such as musty, sour, or fruity, and free from foreign material. Liquid egg products shall be held at a temperature of 40°F or lower and shall not be held for more than 72 hours from the time of pasteurization until the start of formulation of the eggs into product. Frozen whole eggs shall be held at 0°F or lower and used within 120 days. Previously frozen egg products shall be thawed and held at a temperature of 40°F or lower and shall be held for not more than 24 hours prior to use in product.

3.2.10 Starch, freeze-thaw stable. Starch shall be modified waxy maize of the type used in frozen processed foods. The cooked starch slurry shall be bland with essentially no cereal or starchy taste. It shall be white, odorless, and finely pulverized.

3.2.11 Methylcellulose. Methylcellulose shall be a white, free-flowing powder with no off odors and flavors. It shall have a viscosity of 4,000 centipoises in a 2 percent aqueous solution as measured at a temperature of 68°F by a Ubbelohde tube for viscosity measurements. The viscosity shall be certified by a certificate of conformance provided by the manufacturer. Methylcellulose shall comply with the requirements of the Food Chemicals Codex.

3.2.12 Water. Water for formulation and washing shall comply with the National Interim Primary Drinking Water Regulations.

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3.3 Preparation and processing.

3.3.1 Product formulation. The formulas for chocolate, chocolate peppermint, and vanilla pudding powder for oral liquid feeding shall be as follows:

Ingredients	Chocolate Percent by Weight	Chocolate Peppermint Percent by Weight	Vanilla by Percent by Weight
Margarine	3.90	4.00	3.90
Milk, whole, pasteurized	47.19	46.50	48.07
Cream, heavy	14.50	15.50	15.80
Sugar, granulated	13.50	12.50	13.40
Cocoa powder	2.30	2.90	---
Peppermint extract	---	0.50	---
Salt	0.20	0.19	0.20
Vanillin	0.01	0.01	0.03
Eggs, whole	7.40	6.90	7.20
Starch, freeze-thaw stable	3.40	3.40	3.60
Water, cold for starch slurry	3.40	3.40	3.60
Methylcellulose	0.20	0.20	0.20
Water, hot (176°-194°F) for methylcellulose	2.00	2.00	2.00
Water, cold (40°-50°F) for methylcellulose	2.00	2.00	2.00

3.3.1.1 Amount of water. The proportion of water in the formulation is specified to enable pumping of the product. This amount may be increased or decreased to facilitate pumping or to change the viscosity of the product. Changes in the amount of water should be within limits necessary to dissolve the ingredients without unnecessarily increasing the amount of water that must be removed in the freeze dehydration process.

3.3.2 Cooking. The margarine shall be melted and the milk, cream, sugar, salt, and vanillin shall be added. To this mixture shall be added the cocoa for chocolate pudding or the cocoa and the peppermint extract for chocolate peppermint pudding. The mixture shall then be heated to 160°F. A small amount of hot liquid shall be added to the eggs to prevent coagulation, then added to the other ingredients and mixed well. A slurry shall be made with the starch and a small amount of cold water; this slurry shall be added to the

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other ingredients and heated to 180°F. The product shall be held at 180°F for 10 minutes. The heat shall then be turned off and the mixture cooled rapidly to 120°F. Methylcellulose shall be prepared by mixing with hot water (176°-194°F) until all particles are wetted and a smooth paste is obtained. The cold water (40°-50°F) shall be added to this methylcellulose paste, and stirring shall continue until a smooth slurry is obtained. The methylcellulose slurry shall be combined with the other ingredients and stirred until the mixture is smooth. The mixture shall be frozen to 0°F or below within 12 hours after cooking is completed.

3.3.3 Freezing. The product shall be portioned onto freeze dryer trays immediately after cooling and shall be frozen to 0°F or below within 12 hours after cooking is completed. The product shall be freeze dehydrated within 48 hours after being frozen.

3.3.4 Freeze dehydration. The product shall be freeze dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury. The product temperature, as indicated by a suitable instrument, shall be no more than 130°F, and the platen temperature shall be 135°F. After dehydration has been completed, the pressure shall be equalized to atmospheric level with nitrogen, and the product shall be pulverized. During the interim, the product shall be adequately protected from oxygen and moisture by being held either under a nitrogen atmosphere of 2.0 percent oxygen or less or under a vacuum of at least 27.0 inches of mercury for the entire period. If a vacuum is used, it shall be broken with nitrogen. The nitrogen shall comply with U.S. Pharmacopeia grade as specified in the United States Pharmacopeia.

3.3.5 Pulverization. After freeze dehydration, the product shall be pulverized through a U.S. Standard No. 10 screen (0.0787 inch).

3.3.6 Packaging. After pulverization, the product shall be packaged without delay in pouches as specified in MIL-P-44151. In no case shall more than 16 hours elapse from the time the freeze dehydration chamber is opened to the time the product is completely packaged, and at no time shall more than 3 hours elapse from the time the product is pulverized to the time it is completely packaged.

3.4 Finished product requirements.

3.4.1 Dehydrated, pulverized product. The dehydrated product shall comply with the following requirements:

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a. The product shall contain no foreign or extraneous material.

b. The product shall be free from caking and hard lumps.

c. The product shall leave no particles remaining when passed through a U.S. Standard No. 10 sieve.

d. The moisture content shall not exceed 2.0 percent (see 4.5.1).

e. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample (see 4.5.2).

f. The Escherichia coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample (see 4.5.2).

g. The sample average net weight of the lot shall be not less than 80.0 grams.

h. The individual pouches shall contain not more than 88.0 grams nor less than 72.0 grams.

3.4.2 Rehydrated product. Rehydration of the dehydrated powder for oral liquid feeding shall be accomplished by mixing each pouch (80 grams) with 1 cup (8 ounces) of cold water or milk in a high-speed food blender for 30 seconds. For a pureed diet, only 1/2 cup (4 ounces) of liquid should be used. The rehydrated product shall comply with the following requirements:

a. After rehydration for oral liquid feeding, no more than 0.5 gram of product shall remain when the product has passed through a U.S. Standard No. 40 sieve.

b. The rehydrated product shall have a flavor and odor that is characteristic of the specified pudding:

chocolate - chocolate
chocolate peppermint - chocolate peppermint
vanilla - vanilla

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